

# AYUSH GEHLOT

Jersey City, NJ 07306 | [ayush.gehlot@pace.edu](mailto:ayush.gehlot@pace.edu) | 201-275-8024 | [LinkedIn](#) | [HackerRank](#)

## EDUCATION

**Pace University** | New York, NY | Master of Science (MS) in Computer Science | **Concentration:** Software Development May 2026  
**MBM University** | Rajasthan, India | Bachelor of Engineering (BEng) in Electronics and Computer | **GPA:** 7.82/10 June 2023

## RELEVANT COURSEWORK

Object Oriented Programming | Data Structures & Algorithms | Web Technology | Statistics | VLSI Design | Operating Systems | Software Testing | Database Management | Computer Engineering | Information Technology | Cloud Computing | Application Development | Agile Development | AI |

## TECHNICAL SKILLS

**Programming Languages:** C, C++, C#, Java, Python, JavaScript, Node.js, .NET, HTML, CSS, R, VHDL, REST API, Flutter | **Database Management:** MySQL, NoSQL, Oracle, PostgreSQL | **Frameworks:** Vue.js, Angular, Agile, Scrum | **Operating Systems:** Windows, Linux, Android, Mac OS | **Software / Tools:** VS Code, Android Studio, AWS, Microsoft Azure, NetBeans, MS Visio, Postman, Cursor, MS Professional, R Studio, XAMPP, Geany, Tableau, XERO, Qtest, Jira | **Libraries:** REACT, 3Js, GTK

## PROFESSIONAL EXPERIENCE

- The New York Stem Cell Foundation**, New York, NY | Automation/Software Engineer June 2025 – August 2025
- Coordinated with R&D team to research & prototype smart incubator system formulated to preserve human body & blood cells under optimal environmental conditions.
  - Engineered C-based control system utilizing PID algorithms to regulate concentration of oxygen (O<sub>2</sub>), carbon dioxide (CO<sub>2</sub>), nitrogen (N<sub>2</sub>), & internal pressure within incubator.
  - Designed & implemented interactive React-based touchscreen GUI for real-time monitoring & manual configuration of gas levels & environmental parameters on Raspberry Pi platform.
- Dream Team Technologies PVT LTD.**, Rajasthan, India | Full Stack Developer Intern June 2022 – December 2022
- Collaborated in team of Developers & UX/UI Designers to launch enterprise university management platform, creating front-end using HTML, CSS, & JavaScript allowing teachers to enroll scholars, track attendance, & assign assignments.
  - Integrated React, Angular, & Vue.js frameworks to build intuitive user interface, introducing authentication protocols using OAuth to secure payment information & allowing school administrators to maintain records & pull reports.
  - Created automated testing scripts using Selenium UI testing & PyTest, reducing manual testing time by 75% & yielding faster deployment cycles.
- Defense Research & Development Organization (DRDO)**, Rajasthan, India | Hardware Engineering Intern June 2021 – August 2021
- Conducted rigorous validation of VHDL-designed circuits (hardware language like C++) using Model Sim, used to design & simulate digital hardware systems, assessing syntax & structure of VHDL code through its architecture & configuration, achieving 100% verification rate, leading to increased project approvals across teams.

## ACADEMIC PROJECTS & EXPERIENTIAL LEARNING

- Smart City Dashboard — NYC Experiential Learning Lab (Spotify, MongoDB & Kensho Technologies) Hackathon (1<sup>st</sup> Place)** November 2025
- Designed and implemented a **responsive React-based frontend** that rendered **geospatial maps, trend graphs, and forecasting outputs**, enabling non-technical stakeholders to explore city-wide sanitation insights through a clean, intuitive UI.
  - Collaborated with **data engineers, data scientists, and business analysts** to translate **statistical models and predictions into actionable, interactive visualizations** used for operational decision-making.
  - Integrated **MongoDB Atlas** as the **primary data layer** to store, query, and serve large-scale sanitation and complaint datasets, enabling **fast aggregation, filtering, and dashboard-level analytics**.
- AI Restaurant Chatbot for CaterPal Technologies Hackathon (Secured 1<sup>st</sup> Place)** March 2025
- Developed AI-powered chatbot to filter food options based on user input to enhance interaction on partner restaurant sites, integrating Python (Flask) backend with HTML, CSS, JavaScript, MongoDB, & OpenAI API, enabling real-time AI-driven recommendations.
  - Established surplus food donation feature to connect catering services with local charities to address food waste & social impact goals while integrating feedback system to track most recommended dishes & personalize suggestions based on user preferences.
- Advanced Driving Assistant System using Arduino & C++** March 2023 – June 2023
- Synchronized in team of four to develop advanced driving assistant system to support obstacle detection & vehicle navigation, integrating sensors & cameras programmed with image processing to reduce road accidents.
  - Built model car with motor for system testing, creating system with Arduino sensors, LIDAR, & Integrated Radar to detect obstacles with sonic waves, integrating night-vision image sensing to boost detection accuracy & GPS APIs to direct drivers around obstacles, refining prototype to achieve approximately 70% accuracy.
  - Proposed system to India's Minister of Railways, leading to integration of prototype in ministry's systems to improve railway safety.

## LEADERSHIP

- Paws and Wings Society (NGO)**, Founder January 2020 – Present
- Launched community of 60 volunteers & 3 veterinarians to provide medical treatment & shelter to stray animals and birds designed a website to source donations & promote adoptions using React framework, AI assistance, leading to rescue of 4k+ animal lives.